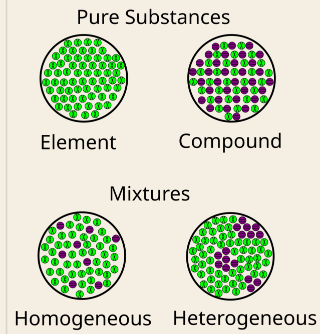
CURIONEST

CHEMISTRY FOUNDATIONS

INTRODUCTION TO MATTER

Complete Instructional Unit for U.S. Grades 9-10



TEACH -> MODEL -> PRACTICE -> ASSESS

- 15 projectable teaching slides
- 12 student lesson, practice, and review pages
- Unit Tests A and B with 20 multiple-choice items each
- Teacher guide, full answers, rationales, and repair moves

Move from what students can see to particle-level claims they can defend.

5 LESSONS · TWO 20-ITEM TEST FORMS · PDF · US LETTER

TEACH

1

Matter at Two Scales

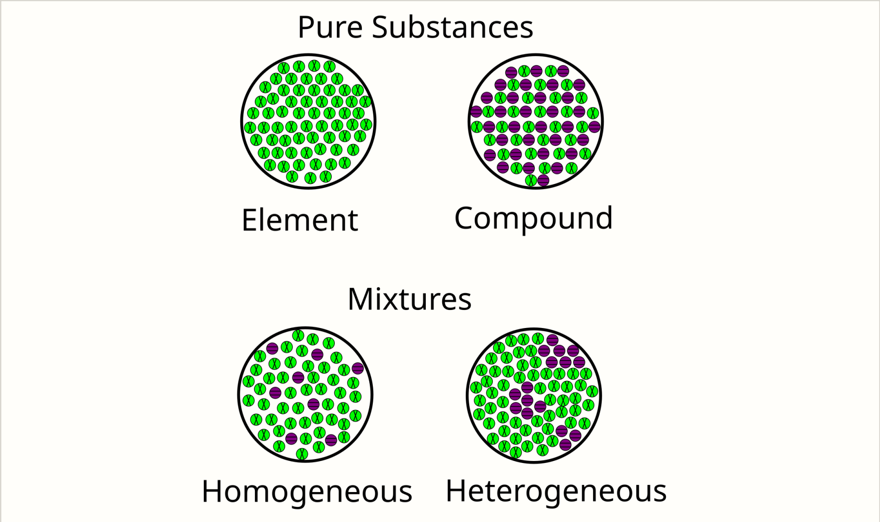
Explain the idea

BIG IDEA

Matter has mass and occupies space. Particle models help explain composition and distribution, but their symbols, colors, spacing, and scale are simplified.

ESSENTIAL TERMS

matter: anything that has mass and occupies space
macroscopic: the scale of observations made directly or with ordinary tools
particle model: a simplified representation of particles and their arrangement
model limitation: a feature of reality that a representation does not show accurately



Particle types and distribution provide model evidence.

WHAT TO REMEMBER

- Use mass and occupied space—not visibility—as the classroom test for matter.
- A particle symbol represents a type of particle; repeated symbols can reveal composition and distribution.
- Identical particles support a pure-substance claim, while more than one particle type supports a mixture claim.
- Models are not to scale and do not prove every property of the real sample.

1 Matter at Two Scales

STUDENT LESSON

Learn and model

Name: ____________________ Date: __________ Period: _____

TARGET **Use the definition of matter and a particle model to make and limit claims about a sample.**

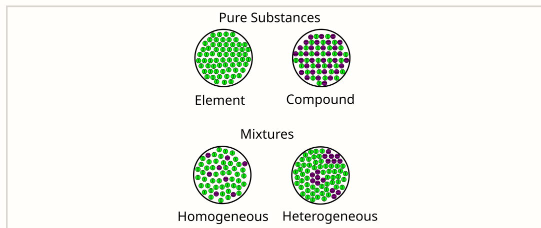
ENGAGE Air is invisible, light is visible, and sound can be heard. Which of the three is matter, and what evidence should decide?

TEACH

- Use mass and occupied space—not visibility—as the classroom test for matter.
- A particle symbol represents a type of particle; repeated symbols can reveal composition and distribution.
- Identical particles support a pure-substance claim, while more than one particle type supports a mixture claim.
- Models are not to scale and do not prove every property of the real sample.

TERMS

matter: anything that has mass and occupies space
macroscopic: the scale of observations made directly or with ordinary tools
particle model: a simplified representation of particles and their arrangement
model limitation: a feature of reality that a representation does not show accurately



Particle types and distribution provide model evidence.

WORKED EXAMPLE

A sealed diagram contains identical two-atom particles. What can be concluded, and what remains unknown?

1. Every represented particle is identical, so the model supports a pure-substance claim.
2. Inspect the atom symbols inside one particle: the same element would indicate an element; different elements would indicate a compound.
3. If the symbols do not distinguish atom identity, the model lacks the evidence needed for the second decision.

RESULT: The sample is modeled as pure. Element versus compound cannot be decided without evidence about the atom types in each particle.

1 Matter at Two Scales

STUDENT PRACTICE

Practice and check

Name: ____________________ Date: __________ Period: _____

GUIDED PRACTICE

L1-1. Classify air in a sealed syringe, visible light, and sound as matter or not matter. Defend each decision.

Scaffold: Apply both parts of the definition: mass and occupied space.

INDEPENDENT

L1-2. In the particle visual, what evidence distinguishes a homogeneous mixture from a heterogeneous mixture?

INDEPENDENT

L1-3. State one useful feature and one limitation of the particle visual.

INDEPENDENT

L1-4. A sample expands to fill its container. Explain why this observation alone does not prove it is not matter.

EXIT TICKET

L1-5. What additional evidence is needed to decide whether identical two-atom particles represent an element or a compound?

Unit Test - Form A

Introduction to Matter

STUDENT ASSESSMENT

Name: ____________________ Date: __________ Period: _____

Directions: Select the best answer for each question. Mark one choice only. Complete this test after all lessons and the cumulative review.

1. Which sample is matter?

A. Visible light

B. Sound from a speaker

C. Air in a sealed syringe

D. A reflected image

2. A particle model contains two particle types evenly distributed. What is the best classification?

A. Element

B. Compound

C. Heterogeneous mixture

D. Homogeneous mixture

3. Which statement describes a limitation of a typical particle model?

A. Particle size and spacing are not shown to scale.

B. It can never show more than one particle type.

C. It proves every property of the sample.

D. Its symbols are literal photographs.

4. Which evidence best supports that a clear solution is a mixture?

A. It is colorless.

B. It occupies a beaker.

C. It has a measurable mass.

D. A physical process recovers two components.

5. How should a clean copper wire be classified?

A. Element

B. Compound

C. Homogeneous mixture

D. Heterogeneous mixture

TEACHER KEY

1

Matter at Two Scales

Practice answers and repair moves

L1-1. Classify air in a sealed syringe, visible light, and sound as matter or not matter. Defend each decision.

ANSWER: Air is matter because it has mass and occupies space. Visible light and sound are forms of energy/waves, not samples of matter in this classroom classification.

L1-2. In the particle visual, what evidence distinguishes a homogeneous mixture from a heterogeneous mixture?

ANSWER: Both have more than one particle type. The homogeneous mixture is uniformly distributed; the heterogeneous mixture has nonuniform regions or clustering.

L1-3. State one useful feature and one limitation of the particle visual.

ANSWER: Useful: it makes composition and distribution visible. Limitation: particle sizes, colors, spacing, and two-dimensional positions are simplified and not to scale.

L1-4. A sample expands to fill its container. Explain why this observation alone does not prove it is not matter.

ANSWER: A gas can expand to fill a container and is still matter because it has mass and occupies space.

L1-5. What additional evidence is needed to decide whether identical two-atom particles represent an element or a compound?

ANSWER: Determine whether the two atoms in each particle are the same element or different elements.

MISCONCEPTION	Only solids and liquids are matter because gases cannot always be seen.
REPAIR	**Use a sealed syringe or inflated balloon as evidence that gas occupies space and contributes mass.**